

Varunprasaath R. Selvaraju, Ph.D.

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PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in analytical characterization, method troubleshooting, DOE-guided process development, and laboratory quality systems. Uses LC-MS, GC-MS, NMR/GPC, FT-IR, UV-Vis-NIR, TGA/DSC, QCM-D, and electrochemistry to support robust methods, investigations, and reproducible material-quality decisions.

CORE COMPETENCIES

LC/GC Method Development

Chromatography Troubleshooting

MSAT Investigations

Method Validation & Transfer

DOE / SPC

Technical Documentation

Materials Characterization

Laboratory Quality Systems

WORK EXPERIENCE

University of Colorado Boulder

July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, converting analytical data and material-quality metrics into reproducible process windows, specifications, and troubleshooting plans.

University of Colorado Boulder

July 2021 - July 2025

Graduate Researcher

- Designed and characterized organic molecular materials across 15+ variants using LC-MS/GC-MS-adjacent workflows, NMR/GPC, FT-IR, UV-Vis-NIR, TGA/DSC, and electrochemistry.

Georgia Institute of Technology

Aug 2019 - July 2021

Graduate Researcher

- Developed thin-film process controls and surface-preparation protocols, linking process conditions to morphology, interface quality, and analytical performance readouts.

ETH Zurich

May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition with DOE and QCM-D monitoring, establishing process-control parameters and repeatable characterization workflows.

University of Colorado Boulder

2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented equipment tracking, calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech

2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Analytical Characterization & Method Robustness Chemistry of Materials 2025

Used orthogonal analytical methods to connect structure, purity, degradation, and material performance across organic material variants.

LC-MS, GC-MS, NMR, GPC, FT-IR, UV-Vis-NIR, TGA/DSC

Bio-Based Polymer Process Controls Current

Defined DOE-guided process windows and material-quality specifications for reproducible polymer scale-up.

DOE, SPC, polymer characterization, technical reports

AFOSR Thin-Film Process Development DoD Funded

Linked CVD/surface preparation conditions to film morphology, interface quality, and optical performance.

CVD, QCM-D, surface prep, thin films

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder

2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials

Advisor: Prof. Seth Marder

(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc)

2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Chromatography & MS: LC-MS, GC-MS, HPLC/GC-adjacent method troubleshooting, sample preparation, NMR, GPC/SEC

Materials & Stability: FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry, degradation/failure analysis, structure-property relationships

Process & Quality Systems: DOE, SPC, JMP (basic), Python (basic), Excel, SOPs, calibration systems, technical reports, reproducibility studies

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." *Chemistry of Materials*, 2025, 37, 5558-5568.

2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." *RSC Advances*, 2018, 8, 20048-20055.

3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.

Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.

Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.

Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)